QCM A · domain and properties

After S47; QCM B after S50.

For each question, only one proposal is accurate.
consult the correction.

1. The domain of $\ln\ (2x-1)$ is:

A. $\mathbb{R}$ B. $[\frac{1}{2};+\infty[$

C. $]0;+\infty[$ D. $]\frac{1}{2};+\infty[$

2. For $a,b>0$, $\ln(ab)$ is valid for:

A. $\ln a+\ln b$ B. $\ln a\ln b$

C. $\ln(a+b)$ D. $a\ln b$

3. $\ln(e^{-3})$ is valid for:

A. 3 B. e^3

C. -3 D. $\ln 3$

4. The derivative of $\ln(1+x^2)$ is:

A. $\frac{1}{1+x^2}$ B. $\frac{2x}{1+x^2}$

C. $2\ln x$ D. $\frac{2}{x}$

5. The equation $\ln x=2$ a for solution:

A. $2e$ B. $\ln 2$

C. 2 D. e^2

QCM B · Threshold, Growth and Parameter

For each question, only one proposal is accurate.
consult the correction.

6. The smallest integer $n\geq 0$ as $(\frac{1}{2})^n<0.1$ is:

A. 3 B. 10

C. 4 D. 2

7. For $0<q<1$ and $0<a<1$, $q^n<a$ equals:

A. $n>\frac{\ln a}{\ln q}$ B. $n<\frac{\ln a}{\ln q}$

C. $n>\frac{\ln q}{\ln a}$ D. $n>a/q$

8. $\lim_{x\to+\infty}\frac{\ln x}{x}$ is valid for:

A. 1 B. 0

C. $+\infty$ D. $-\infty$

9. $\ln(x^2)=2\ln x$ is valid:

A. for all real x B. for everything $x\neq 0$

C. only for $x>1$ D. for everything $x>0$

10. The maximum of $g(x)=\frac{\ln x}{x}$, $x>0$, is:

A. 1 B. e

C. $1/e$ D. 0

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. $\ln(1+1)=\ln 1+\ln 1$.

2. Divide inequality by ln 2 reverses its meaning.
3. $\ln x / x = 1$ has a positive solution.